
Write all the questions:

4*10=40

1Q File Transfer & Integrity Verification (10 m)

- I. On **VM1 (Legacy Linux, IP: 10.11.22.33)**, create a file named **config_v1.txt** containing **10–15 lines** of configuration data or text.
 - II. Transfer this file to **VM2 (bodhi linux IP: 192.0.5.5)** using a **command-line method** (scp or nc).
 - III. Update the file on **VM1** (create **config_v2.txt** with additional 5 lines), and **transfer it again** to VM2 without overwriting the previous version.
 - IV. On **VM2**, verify that both versions (**config_v1.txt** and **config_v2.txt**) exist and display their contents.
- -----

2Q. set up bodhi linux inside virtualbox (10 Marks)

2.1 Write a C program to generate Fibonacci series

- i. Compile using gcc filename.c
- ii. Save output in a text file

2.2 Create a directory named cc_lab in your home directory.

- iii. Execute prime number code

2.3 Compress a folder called project into project.tar.gz

2.4 Create a folder exam_lab

Inside it, create 5 text files and list them

Move selected files into another directory

3Q CloudSim Simulation — Multi-VM Arithmetic Scheduling (10 Marks)

Write a CloudSim program with the following requirements:

1. Environment Setup

- **Create:**
 - Three Datacenter
two DatacenterBroker
four Virtual Machines (VMs)
 - At least 5 Cloudlets

2. Logic

- Each Cloudlet should have different lengths.
Assign Cloudlets alternately to VM1 and VM2 (round-robin assignment).
Inside the program, use **switch-case** to perform arithmetic operations:
- Cloudlet ID 0 → Addition
Cloudlet ID 1 → Subtraction
- Cloudlet ID 2 → Multiplication
Cloudlet ID 3 → Division
Cloudlet ID 4+ → Modulus

Output

After simulation ends, display for each Cloudlet:

Cloudlet ID | VM ID | Operation | Result | Execution Status

4Q. ALL READY U Install NGINX on Linux and configure it to serve a static HTML homepage.

Output Required: Homepage should open in browser using server Ip.

4.1 Create a Python Flask Application that displays:

- **User Name**
- **System IP Address**
- **Current date & time**

Expose the Flask app on port 5000.

4.2 Configure NGINX Reverse Proxy:

- When user opens `http://server_ip/app`, request should go to Flask backend (port 6000)

GI to run Flask as a background service.

Check: Service persists after closing the terminal.